

Report Period	Sep 26, 2026 20:02 - Sep 26, 2026 20:09
Generated At	2026-09-26 20:09:00
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		32894	
Total Threats Mitigated	29	Total Packets Analyzed		2648172856	
High Priority	3	Malicious Packets		2639893231	
Low Priority	26	True Negatives Passed		26505	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		16	
False Positives	237	UDP Flood		3	
FP Rate	0.72%	Uncertain		10	
IF Precision	97.02%				
IF Recall	94.38%	Action		Count	
IF F1-Score	95.68%	Quarantined		5	
IF Accuracy	98.30%	Time Ban		4	
		Blackhole		0	
		Released		7	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	97.02%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.38%	Share of actual attacks successfully flagged
F1-Score	95.68%	Balanced measure combining Precision and Recall
Accuracy	98.30%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.72%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.62%	Actual attacks that went undetected
True Positive Rate (TPR)	94.38%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.28%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6176	FN 368
Actual: Normal	FP 190	TN 26137

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.84%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.92%	Balanced measure combining Precision and Recall
Accuracy	99.84%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	5.26%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	94.74%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	6176	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	97.02%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.38%	Share of actual attacks successfully flagged
F1-Score	95.68%	Balanced measure combining Precision and Recall
Accuracy	98.30%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.72%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.62%	Actual attacks that went undetected
True Positive Rate (TPR)	94.38%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.28%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6176	FN 368
Actual: Normal	FP 190	TN 26137

2d. Response Latency

Metric	Value	Description
Detection Time	539.3 ms	Average time taken to detect an attack
Mitigation Response Time	1.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.67%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	9.51%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	8.28%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.9	4	4	2026-09-26 20:06:19	2026-09-26 20:08:53
10.0.0.16	4	3	2026-09-26 20:05:53	2026-09-26 20:08:33
10.0.0.17	4	3	2026-09-26 20:06:00	2026-09-26 20:08:40
10.0.0.18	4	3	2026-09-26 20:05:54	2026-09-26 20:08:35
10.0.0.19	4	3	2026-09-26 20:05:54	2026-09-26 20:08:35
10.0.0.4	5	1	2026-09-26 20:06:13	2026-09-26 20:07:06
10.0.0.1	1	1	2026-09-26 20:06:35	2026-09-26 20:06:45
10.0.0.10	1	1	2026-09-26 20:06:14	2026-09-26 20:06:25

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.2	1	1	2026-09-26 20:06:43	2026-09-26 20:06:53
10.0.0.3	1	1	2026-09-26 20:06:38	2026-09-26 20:06:48
10.0.0.5	1	1	2026-09-26 20:06:18	2026-09-26 20:06:29

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 20:05:53	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:05:54	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:05:54	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:06:00	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:06:03	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 20:06:04	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 20:06:04	10.0.0.19	DDoS	SYN Flood	90.0%	Critical	Time Ban (30s)
2026-09-26 20:06:04	10.0.0.18	DDoS	SYN Flood	90.0%	Critical	Time Ban (30s)
2026-09-26 20:06:05	10.0.0.18	DDoS	SYN Flood	92.8%	Critical	Time Ban
2026-09-26 20:06:05	10.0.0.19	DDoS	SYN Flood	93.1%	Critical	Time Ban
2026-09-26 20:06:10	10.0.0.17	DDoS	SYN Flood	90.1%	Critical	Time Ban (30s)
2026-09-26 20:06:10	10.0.0.17	DDoS	SYN Flood	90.1%	Critical	Time Ban
2026-09-26 20:06:19	10.0.0.9	DDoS	UDP Flood	92.4%	High	Time Ban (30s)
2026-09-26 20:06:24	10.0.0.4	DDoS	Uncertain	60.6%	Low	Released
2026-09-26 20:06:25	10.0.0.10	DDoS	Uncertain	68.7%	Low	Released
2026-09-26 20:06:29	10.0.0.5	DDoS	Uncertain	60.1%	Low	Released
2026-09-26 20:06:45	10.0.0.1	DDoS	Uncertain	61.4%	Low	Released
2026-09-26 20:06:48	10.0.0.3	DDoS	Uncertain	61.4%	Low	Released
2026-09-26 20:06:49	10.0.0.9	DDoS	UDP Flood	92.4%	High	Released
2026-09-26 20:06:53	10.0.0.2	DDoS	Uncertain	61.2%	Low	Released
2026-09-26 20:07:04	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-26 20:07:05	10.0.0.19	DDoS	SYN Flood	93.9%	Critical	Time Ban (1m)
2026-09-26 20:07:05	10.0.0.18	DDoS	SYN Flood	93.7%	Critical	Time Ban (1m)
2026-09-26 20:07:11	10.0.0.17	DDoS	SYN Flood	93.0%	Critical	Time Ban (1m)
2026-09-26 20:07:19	10.0.0.9	DDoS	UDP Flood	92.4%	High	Quarantined
2026-09-26 20:08:33	10.0.0.16	DDoS	SYN Flood	93.1%	Critical	Time Ban (2m)
2026-09-26 20:08:35	10.0.0.19	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-26 20:08:35	10.0.0.18	DDoS	SYN Flood	90.1%	Critical	Time Ban (2m)
2026-09-26 20:08:40	10.0.0.17	DDoS	SYN Flood	93.2%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.9	UDP Flood	0.6118	92.4%	Critical	Phase 1	4	10s	2026-09-26 20:08:53	Attack Stopped
10.0.0.17	SYN Flood	0.6423	93.2%	Critical	Phase 2	1	10s	2026-09-26 20:08:40	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6443	90.1%	Critical	Phase 2	1	11s	2026-09-26 20:08:35	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6516	93.4%	Critical	Phase 2	1	11s	2026-09-26 20:08:35	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6380	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:08:33	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6102	91.0%	High	Phase 1	3	11s	2026-09-26 20:08:09	Attack Stopped
10.0.0.9	UDP Flood	0.6118	92.4%	High	Phase 1	2	11s	2026-09-26 20:07:30	Attack Stopped
10.0.0.17	SYN Flood	0.6345	93.0%	Critical	Phase 2	3	1m 11s	2026-09-26 20:07:11	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6130	60.6%	Low	Phase 1	1	10s	2026-09-26 20:07:06	Attack Stopped
10.0.0.18	SYN Flood	0.6062	93.7%	Critical	Phase 2	3	1m 11s	2026-09-26 20:07:05	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6253	93.9%	Critical	Phase 2	3	1m 11s	2026-09-26 20:07:05	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6271	92.9%	Critical	Phase 2	3	1m 11s	2026-09-26 20:07:04	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6163	60.6%	Low	Phase 1	1	11s	2026-09-26 20:06:55	Attack Stopped
10.0.0.2	Uncertain	0.5852	61.2%	Low	Phase 1	1	10s	2026-09-26 20:06:53	Attack Stopped
10.0.0.9	UDP Flood	0.5569	92.4%	High	Phase 2	1	30s	2026-09-26 20:06:49	Ban Expired
10.0.0.3	Uncertain	0.5925	61.4%	Low	Phase 1	1	10s	2026-09-26 20:06:48	Attack Stopped
10.0.0.1	Uncertain	0.5900	61.4%	Low	Phase 1	1	10s	2026-09-26 20:06:45	Attack Stopped
10.0.0.4	Uncertain	0.6156	60.6%	Low	Phase 1	1	10s	2026-09-26 20:06:44	Attack Stopped
10.0.0.17	SYN Flood	0.6391	90.9%	Critical	Phase 2	2	41s	2026-09-26 20:06:41	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6414	93.6%	Critical	Phase 2	2	41s	2026-09-26 20:06:35	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6458	93.9%	Critical	Phase 2	2	41s	2026-09-26 20:06:35	Attack persisted - escalating to time ban
10.0.0.4	Uncertain	0.6139	60.6%	Low	Phase 1	1	10s	2026-09-26 20:06:34	Attack Stopped
10.0.0.16	SYN Flood	0.6324	92.9%	Critical	Phase 2	2	41s	2026-09-26 20:06:34	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.5917	60.1%	Low	Phase 1	1	11s	2026-09-26 20:06:29	Attack Stopped

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.10	Uncertain	0.5632	68.7%	Low	Phase 1	1	11s	2026-09-26 20:06:25	Attack Stopped
10.0.0.4	Uncertain	0.6065	60.6%	Low	Phase 1	1	11s	2026-09-26 20:06:24	Attack Stopped
10.0.0.17	SYN Flood	0.6414	90.1%	Critical	Phase 2	1	10s	2026-09-26 20:06:10	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6473	90.0%	Critical	Phase 2	1	10s	2026-09-26 20:06:04	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6517	90.0%	Critical	Phase 2	1	10s	2026-09-26 20:06:04	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6396	92.9%	Critical	Phase 2	1	10s	2026-09-26 20:06:03	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE